

ASSIGNMENT

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CSA5117 - Cryptography

and

Network Security

SET-2

1Q: Blowfish - S-Box 1 in F Function

Given i/p:

01101001

01101001

01101001

Each 01101001 is;

$$01101001_2 = 105_{10}$$

In Blowfish, the input to the F-function is normally 32 bits, divided into four 8-bit parts:

$$X = a || b || c || d$$

The first 8 bits are used as the index for S-box 1

so,

$$a = 01101001 = 105$$

$$S_1[105]$$

The F-function then uses all four S-boxes

$$F(x) = [(S_1[a] + S_2[b]) \oplus S_3[c] + S_4[d]]$$

2) Diffie-Hellman key exchange

Given;

$$g=5, p=23$$

Alice's private key:

$$a=6$$

Bob's private key:

$$b=15$$

Alice public key

$$A = g^a \bmod p$$

$$A = 5^6 \bmod 23 = 8$$

Bob's public key

$$B = g^b \bmod p$$

$$B = 5^{15} \bmod 23 = 19$$

Shared key

Alice calculates:

$$K = B^a \bmod p$$

$$K = 19^6 \bmod 23 = 2$$

Bob calculates:

$$K = A^b \bmod p$$

$$K = 8^{15} \bmod 23 = 2$$

shared key = 2

3) Blowfish vs RC5 vs DES

Feature	Blowfish	RC5	DES
Key Size	32-448 bits	Variable	56 effective bits
Structure	Feistel	Variable structure	Feistel
Rounds	16	variable	16
Block Size	64 bits	32/64/128 bits	64 bits
Performance	Fast	Fast and Flexible	slower

4) Block cipher vs stream cipher

Block cipher

- Encrypts data in fixed-size blocks.
- Examples :- AES, Blowfish, DES
- used for files, databases etc---

Example

A 128-bit block of a file -> encrypted

-> ciphertext block.

Stream cipher

- Encrypts data continuously, usually bit-by-bit or byte-by-byte.
- Example: ChaCha20
- useful for real time communication

Basic formula

$$C = P \oplus K$$

Where,

- P = plaintext
- K = keystream
- C = cipher text

Simple difference

Blocks = fixed blocks

stream = Continuous data

5) Importance of Key Management in
Public-key Cryptography

→

• public-key systems use:

→ public key: can be shared

→ private key: must be kept secret.

why key management is important?

1) protects private keys from the attackers.

2) ensures the correct public keys that belongs to the correct person / website.

3) prevents impersonation

4) Allows keys to be renewed or revoked when compromised.

→ That is why key management is

Public Key Cryptography is important.

Digital Certificate

A digital certificate connects:

Identity + public key

A certificate authority (CA) verifies the identity and digitally signs the certificate.

Example

When visiting a banking website, the browser checks its certificate to verify ~~that~~ the public key really belongs to that website.

-> If a private key is stolen, the

Certificate can be revoked.

Main point

Good Key Management = &

confidentiality + authentication

+ security.

Q